

Introduction to Robotics, Lecture 7

Mobile robots

Ways of moving

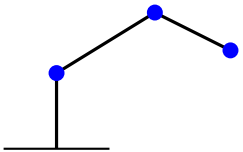
Kinematics of wheeled bases

MIPT

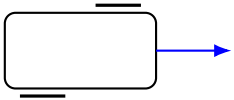
Center for Cognitive Modeling

A mobile robot has to push against the world

Manipulation	Locomotion
q: joint angles; FK: forward kinematics. TCP: tool centre point. Joint angles give the TCP position and orientation.	$\dot{\varphi}$: wheel angular speed. Wheel speeds give robot velocity. Integrating velocity gives position and orientation (pose).



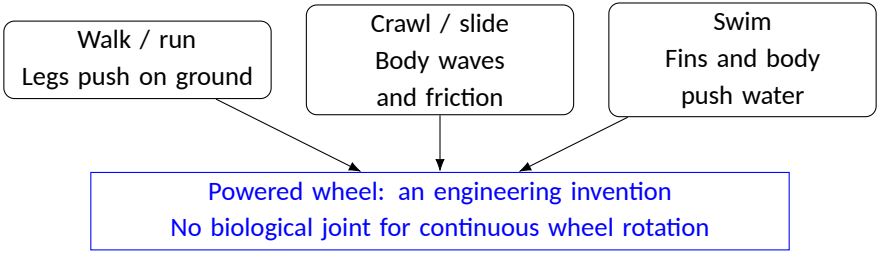
Base bolted to the world



The base carries itself

Every move is a push against something. What do animals push against?

Nature moves in many ways, but never on wheels



Why no wheel joint?

A driven wheel needs unlimited rotation and a way to supply power across the joint. Living limbs use a different arrangement.

What does a wheel do better, and how is it related to a leg?

A wheel is the limit of a many-legged walker

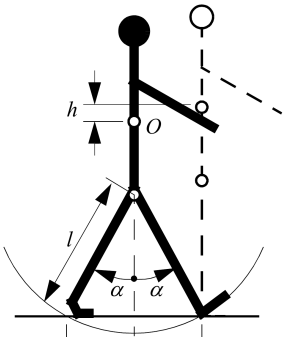


Figure: A rolling polygon approaches a wheel as $d \to 0$.

Centre O; radius l; step d; vertical excursion h; α: half the angle between supports. Which should we choose?

Three questions for any way of moving

1. Stability

Contact number and geometry; centre of mass. Example: a narrow base can tip.

2. Contact

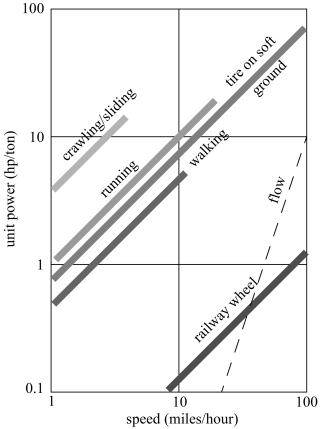
Contact shape, angle and friction. Example: a tyre needs traction to push.

3. Environment

Structure, hardness and medium. Example: sand deforms; a hard floor does not.

Which matters most for a lunar rover? For a vacuum cleaner?

Wheel or leg: which is more efficient?



Power per unit mass at each speed

On hard, flat ground the rail wheel needs 1–2 orders of magnitude less power than walking or running; soft ground raises tyre losses.

Power per mass divided by speed is specific resistance: energy per weight and distance.

Why build legs?

Legs choose where to touch the ground

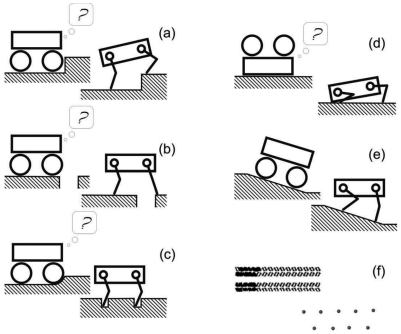


Figure: Legs use separate footholds.

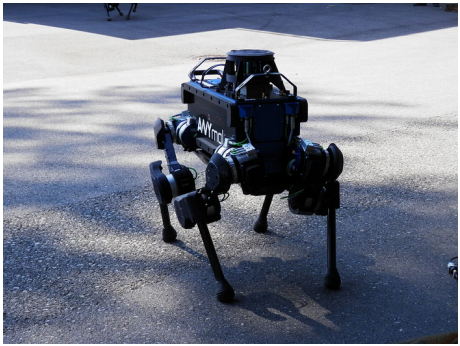


Figure: A quadruped robot on rough ground.

Legs can level the body and recover after a fall; wheels need continuous support.

What do these capabilities cost?

What legs cost

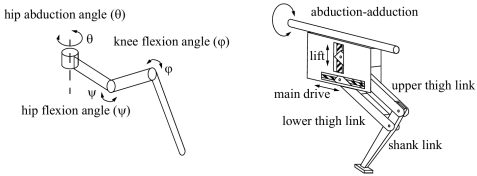


Figure: Two legs with three independent joint motions each.

How many legs are needed just to stand?

Degrees of freedom (DoF)

DoF: one independent joint motion. Lifting and swinging usually needs two; three improve placement.

- More actuators: more mass and control.
- Legs must accelerate their own mass.
- Contact and balance need coordination.

How many legs are needed to stand?

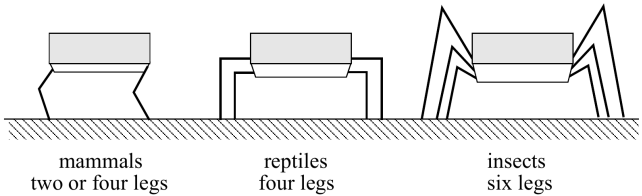
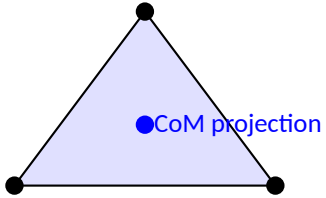


Figure: Two, four and six legs.



Support polygon: the convex hull of ground contacts. CoM means centre of mass.

With point contacts on level ground, three non-collinear contacts can support static balance. Four legs let one lift; six can alternate supporting tripods. Finite feet can provide a wider support region.

Is staying inside that region the only way to balance?

Static, quasi-static and dynamic walking

Static

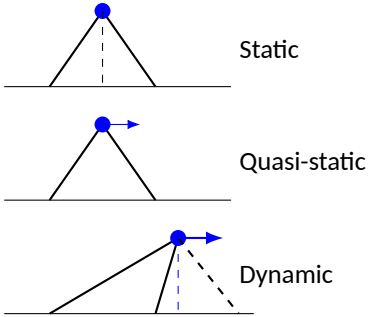
Freeze it: it stays up. The CoM projection stays in the support polygon.

Quasi-static

Move slowly; neglect inertia. Keep the CoM projection in the support polygon.

Dynamic

Freeze it: it falls. Motion and the next landing maintain balance.



Tripod hexapod: static. Atlas and MIT Cheetah: dynamic. Dynamic movement enables running and jumping.

Video break: static versus dynamic

Hexapod Walking Test – Tripod, Wave & Ripple Gaits

https://www.youtube.com/watch?v=RLxne70q6Fo

How many legs touch the ground?

Steve Collin's Passive Dynamic Robot

https://www.youtube.com/watch?v=e2Q2Lx8O6Cg

No motors: watch it walk down a slope.

Parkour Atlas

https://www.youtube.com/watch?v=LikxFZZO2sk

When are both feet off the ground?

How is the order of leg movements chosen?

Gait: the order in which legs move

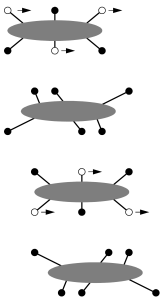


Figure: Alternating support tripods.

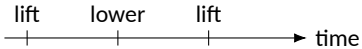
Which orders are useful, and why?

Gait: a repeating sequence of foot lift and lower events.

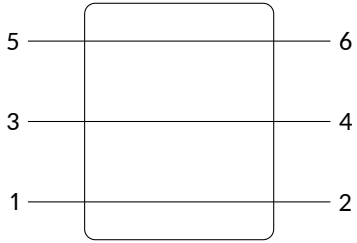
For k legs, count cyclic event orders:

$$N = (2k-1)!$$

$k = 2$: $N = 6$.
$k = 6$: $N = 11! = 39\,916\,800$.
Event orders need not be useful or stable.

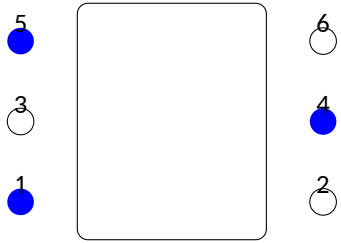


Why different gaits exist: speed against stability



Wave: $1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 5 \rightarrow 6$

Five feet support the body while one moves: most stable, slowest.



Tripod: $\{1, 4, 5\} \leftrightarrow \{2, 3, 6\}$

Fastest of the three static hexapod gaits; ripple lies between wave and tripod.

Quadruped trot alternates diagonal pairs; bound alternates front and rear pairs. Speed, terrain and energy favour different gaits. What makes each step expensive, and how can legs recover energy?

Why walking is expensive and how legs save energy

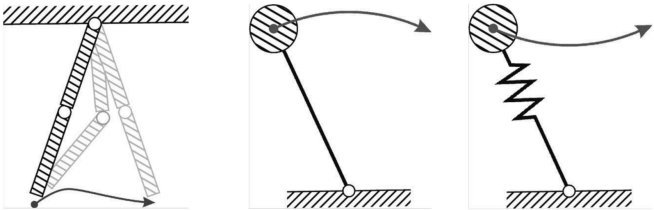


Figure: Swing: double pendulum. Stance: inverted pendulum. Running: spring.

Cost of transport: energy per weight and distance (dimensionless). Each step accelerates and brakes legs; negative work is not fully recovered. Passive walkers use gravity and body dynamics; springs return landing energy.

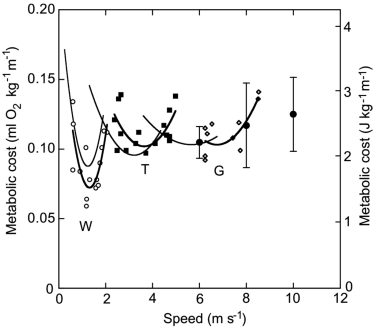


Figure: Metabolic cost: walk (W), trot (T), gallop (G).

Who first used these ideas?

History 1: Raibert's hopper (1983)

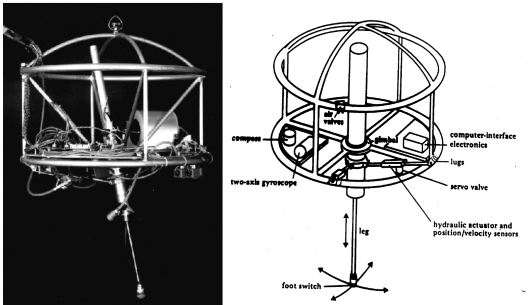


Figure: The Raibert hopper.

Can legs instead be simple and robust?

One leg, active balance

The robot cannot stand statically on a point foot. It continually corrects body attitude and speed by changing the leg angle.

Hydraulic extension produces the next hop; an off-board pump supplies power. Forward speed is controlled through foot placement.

History 2: RHex, simple legs that rotate (2001)



Figure: RHex, a six-legged robot.

Mechanical simplicity

Six rotating legs; one actuator per leg. Alternating tripods produce rough-terrain motion.

Research behaviours ran mostly open-loop, on a clock with little sensing. Toe slip on sand and rubble limits prediction.

No clear single reason for its commercial follow-up history is established; a commercial version did exist.

What changed with more articulated, actively controlled quadrupeds?

History 3: BigDog to Spot





Figure: BigDog (2005): gasoline engine and hydraulics, about 150 kg payload; ended 2013 as too noisy.



Figure: Spot: electric and quieter, about 20 kg payload; sold from 16 June 2020.

Engine-driven hydraulics brought noise and practical limits; Spot uses quieter electric drive. Quadrupeds are one branch. What about robots shaped like us?

History 4: humanoids

ASIMO

Unveiled in November 2000; development stopped in 2018; retired from performing in March 2022. Controlled around a zero moment point (ZMP) criterion: balanced in motion.

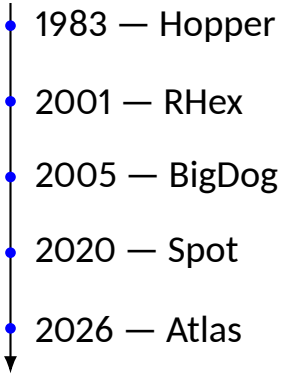
Atlas

Hydraulic Atlas: 2013. Electric Atlas: April 2024. Product version: 5 January 2026 at CES, the Consumer Electronics Show.

A human-shaped body raises the same balance and contact problems. Dynamic motion makes flight phases and controlled landings possible.

What documented changes connect these machines?

Why the field moved: what changed



Documented changes: actuation

Engine-driven hydraulics to electric drive: less noise.

Documented changes: control

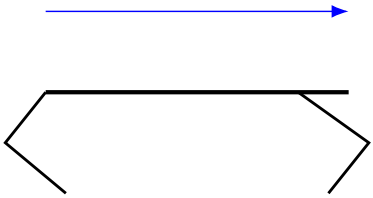
Clock-driven behaviours to feedback-controlled dynamic motion.

Documented changes: commercial products

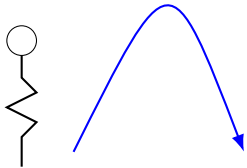
Spot in 2020; product Atlas in 2026.

What else can robots learn from animals?

Biomimetic robots 1: running and jumping



Running and obstacle clearance

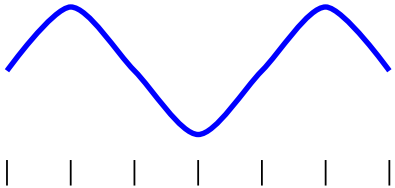


Spring-assisted jumping

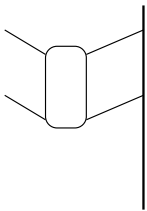
MIT Cheetah (2015)	Salto (2016)
Electric actuators; onboard laser ranging. Ran and jumped autonomously over obstacles up to about 46 cm at about 8 km/h.	Galago-inspired; 100 g, 26 cm. Spring energy enables repeated jumps about 1 m high.

Not all animals walk. What about crawling and climbing?

Biomimetic robots 2: crawling and climbing



Articulated body waves

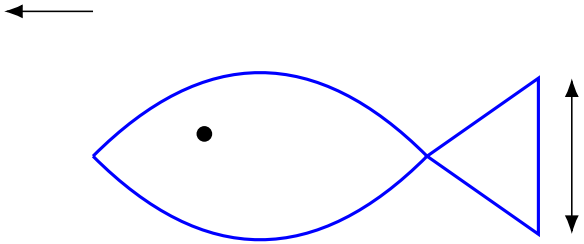


Adhesion and controlled foot forces

CMU modular snake	Stickybot
Carnegie Mellon University: tethered, 16 modules; undulation and rolling. Deployed in a Mexico City earthquake building (September 2017); no survivors found.	Gecko-inspired directional dry adhesive. Layered compliance, directional adhesion and foot-force control; climbed glass and glazed tile at 4 cm/s.

How does the environment change the design in water?

Biomimetic robots 3: swimming

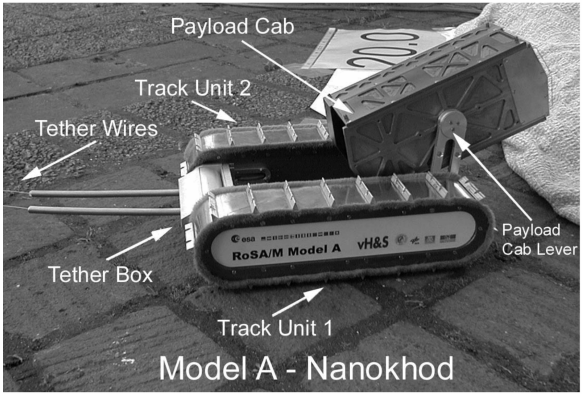


SoFi: a soft robotic fish (2018)	Return to the three questions
Undulates its tail to swim. Adjustable weight and a buoyancy control unit regulate depth.	Water supports weight and provides reaction forces. Contact with a floor is no longer required.

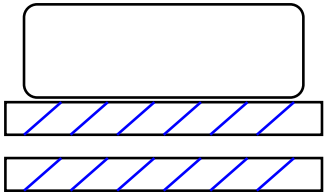
Tested at Rainbow Reef, Fiji: 40-minute dives to over 50 ft (about 15 m), near real fish without scaring them.

Back on land, what can tracks and screws offer?

Tracks and screws



Can we combine the advantages of wheels and legs?



Screw drums

Nanokhod is a tracked microrover. Tracks spread load over a large patch; turning requires skid. Terrain friction reduces odometry accuracy.

Wheel-leg hybrids

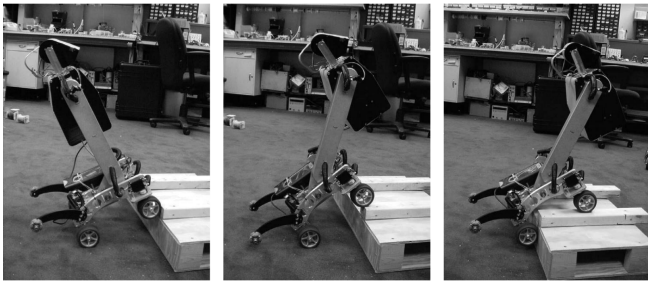


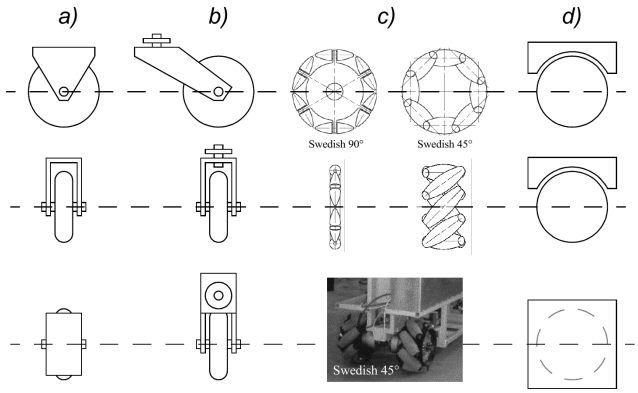
Figure: Personal Rover shifts its centre of mass to climb.

Wheels dominate on flat ground. What wheel types exist?

Mechanism	Main trade-off
Legs	Footholds; swing losses
Tracks	Large patch; skid turns
Screws	Helical contact
Wheels	Efficient; continuous support
Hybrids	Adaptation; complexity

Rolling on level ground; weight shifting for ledges.

Wheel types



How are wheels arranged on a robot?

Four wheel types

Standard: rolls forward; resists sideways motion. Fixed or actively steered.

Castor: passive steering; axis offset from contact.

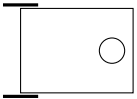
Swedish: free rollers add a direction of motion.

Spherical: a ball rolls in any direction.

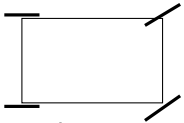
Wheel arrangements



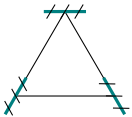
Tricycle
fixed pair + steered



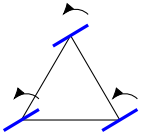
Differential
+ castor



Ackermann
linked steering



Swedish
free rollers



Synchro drive
linked drive / steering

Differential: independent wheel speeds. Ackermann: linked front steering. Tricycle: one steered wheel. Omni: lateral motion. Synchro: linked drive and steering. Which keeps contact on uneven ground?

Stability on uneven ground





Figure: Cye: two wheels, low centre of mass.

Two wheels: CoM below axle. Three contacts: support triangle. More rigid supports need suspension on uneven ground.

How do the locomotion choices compare?



Figure: A rocker-bogie suspension model.

Part A summary and the bridge

	Stability	Contact	Environment	Energy
Legs	Gait dependent	Chosen footholds	Gaps, steps	Swing losses
Tracks	Broad support	Large patch	Soft ground	Turning losses
Wheels	Support geometry	Rolling	Hard, flat	Efficient
Hybrids	Adaptable	Mixed	Uneven	Compromise

From design choice to kinematics

On flat ground wheels win. Now we need a language to say where a wheeled robot goes when its wheels turn.

Where is the robot? We need position and orientation.

Pose and two coordinate frames

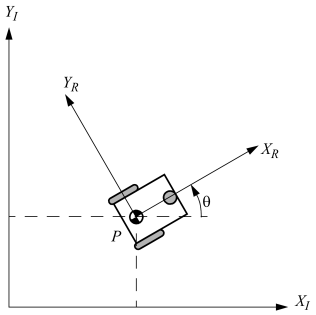


Figure: The global and robot coordinate frames.

How do velocities convert between these frames?

A **coordinate frame** is an origin plus axes.

Global frame $\{X_I, Y_I\}$: fixed to the world; I means inertial.

Robot frame $\{X_R, Y_R\}$: attached at reference point P.

$$\xi_I = \begin{bmatrix} x & y & \theta \end{bmatrix}^{\mathsf{T}}$$

Pose ξ_I: position (x, y) of P and heading θ in the global frame. T denotes transpose.

Converting between the two frames

$$\dot{\xi}_R = R(\theta)\dot{\xi}_I$$

$$R(\theta) = \begin{bmatrix} \cos\theta & \sin\theta & 0 \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\dot{\xi}_I = R(\theta)^{-1}\dot{\xi}_R = R(\theta)^{\mathsf{T}}\dot{\xi}_R$$

A dot means rate of change. $\dot{\xi}_I$ and $\dot{\xi}_R$ are the same velocity expressed in global and robot components. $R(\theta)$ rotates those components.

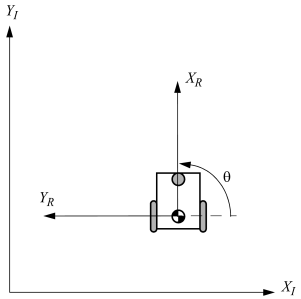
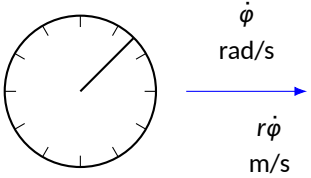


Figure: At $\theta = \pi/2$, robot forward is global $+Y_I$.

Check: $\dot{x}_R = 1$, $\dot{y}_R = 0$ gives $\dot{x}_I = 0$, $\dot{y}_I = 1$.

What the robot measures and what we want



Encoder: counts shaft rotation

Encoder

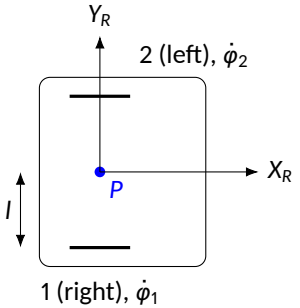
A shaft sensor that counts ticks as the wheel turns. Changes in counts give how far and how fast it turns.

r: wheel radius. $\dot{\varphi}$: wheel angular speed. Under pure rolling, $r\dot{\varphi}$ is the wheel-centre speed along the wheel plane.

Chassis forward kinematics maps wheel speeds to robot velocity. Odometry integrates that velocity into pose; wheel sensors alone do not directly measure global pose.

Do this for the simplest chassis: two driven wheels.

Differential drive: each wheel adds its own part



$$\dot{x}_R = \frac{r}{2}(\dot{\varphi}_1 + \dot{\varphi}_2), \quad \dot{y}_R = 0$$

$$\dot{\theta} = \frac{r}{2l}(\dot{\varphi}_1 - \dot{\varphi}_2)$$

$$\dot{\xi}_I = R(\theta)^{-1} \begin{bmatrix} r(\dot{\varphi}_1 + \dot{\varphi}_2)/2 \\ 0 \\ r(\dot{\varphi}_1 - \dot{\varphi}_2)/(2l) \end{bmatrix}$$

l: half the distance between wheels.

r: wheel radius. $\dot{\varphi}_1$: right wheel; $\dot{\varphi}_2$: left wheel angular speed.

See the same command on three bases.

Equal speeds: straight. Opposite speeds: spin in place. If wheel 1 is faster, which way does the robot turn?

Demo 1: one command, three robots, then noise

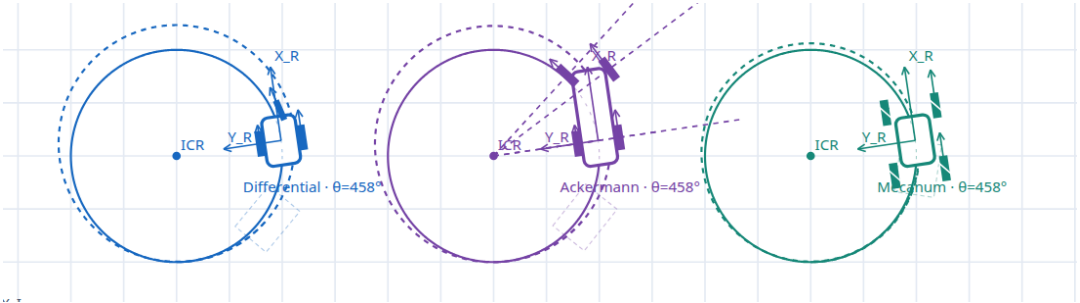


Figure: True paths (solid) and odometry (dashed), with encoder noise.

Set $v = 0.4$ m/s, $\omega = 0.4$ rad/s, then enable encoder noise. Odometry = pose from integrating wheel speeds; errors accumulate. What does a wheel forbid?

What a wheel forbids

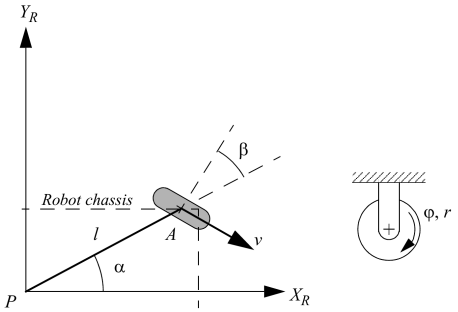


Figure: A fixed wheel at point A, relative to reference point P.

Write both requirements as equations.

$l = |PA|$: distance from P to the wheel. α: angle of PA from X_R. β: wheel-axis angle relative to PA. r: wheel radius; φ: wheel rotation.

Rolling: velocity along the wheel plane matches $r\dot{\varphi}$.

No sideways slip: the lateral contact velocity is zero. Under pure rolling, the material contact point is instantaneously stationary relative to the ground.

The two equations of a fixed wheel

Rolling along the wheel plane

$$\begin{bmatrix}\sin(\alpha+\beta) & -\cos(\alpha+\beta) & -l\cos\beta\end{bmatrix} R(\theta)\dot{\xi}_I - r\dot{\varphi} = 0$$

The chassis velocity at the wheel matches its rolling speed.

No sliding across the wheel plane

$$\begin{bmatrix}\cos(\alpha+\beta) & \sin(\alpha+\beta) & l\sin\beta\end{bmatrix} R(\theta)\dot{\xi}_I = 0$$

The wheel cannot move sideways under the no-slip assumption.

How do other wheel types change these constraints?

The other wheels in one table

Wheel	Fixed / free	Rolling	Sideways constraint
Steered	Controlled $\beta(t)$	Same form	Same form at current angle
Castor	Free swivel; offset	Same form	Extra swivel-rate term
Swedish	Free roller	Roller angle enters	Roller motion absorbs lateral speed
Spherical	Free rolling direction	Direction follows motion	No chassis restriction

Castor: d is steering offset; $\dot{\beta}$ is free swivel rate and can satisfy the sideways equation.

$$\begin{bmatrix}\cos(\alpha+\beta) & \sin(\alpha+\beta) & d+l\sin\beta\end{bmatrix} R(\theta)\dot{\xi}_I + d\dot{\beta} = 0$$

How are the constraints of several wheels combined?

From wheels to the robot

Stack the rolling equations

$$J_1(\beta_s)R(\theta)\dot{\xi}_I - J_2\dot{\varphi} = 0$$

J_1: rolling projections; J_2: diagonal wheel radii; $\dot{\varphi}$: wheel-rate vector.

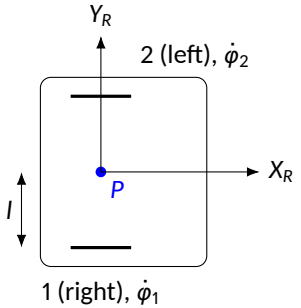
Stack the no-sideways-slip equations

$$C_1(\beta_s)R(\theta)\dot{\xi}_I = 0$$

C_1: sliding constraints; β_s: steering angles. Fixed rows are constant; steered rows vary.

Use the equations on the differential drive first.

Example: differential drive



ICR: instantaneous rotation centre. If $\omega \neq 0$, $\mathrm{ICR}_R = (0, v/\omega)$, on the common axle.

What changes when each wheel has a free roller?

Two rolling equations; one independent lateral constraint:

$$\begin{bmatrix} 1 & 0 & l \\ 1 & 0 & -l \\ 0 & 1 & 0 \end{bmatrix} \dot{\xi}_R = \begin{bmatrix} r\dot{\varphi}_1 \\ r\dot{\varphi}_2 \\ 0 \end{bmatrix}$$

$$\dot{\xi}_R = \begin{bmatrix} r(\dot{\varphi}_1 + \dot{\varphi}_2)/2 \\ 0 \\ r(\dot{\varphi}_1 - \dot{\varphi}_2)/(2l) \end{bmatrix}$$

For straight travel, the ICR is at infinity.

Example: omnidirectional robot (three Swedish wheels)

C_1 empty; J_1 invertible: $\dot{\xi}_I = R(\theta)^{-1} J_1^{-1} J_2 \dot{\varphi}$.

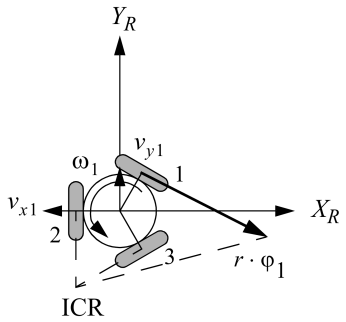


Figure: Three free velocity components.

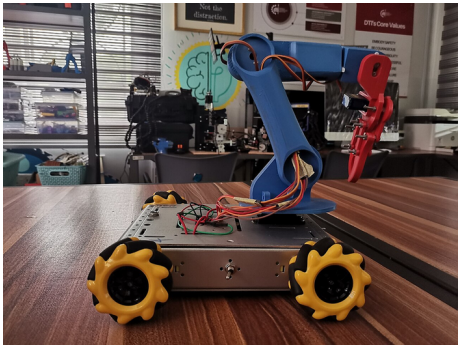


Figure: A mecanum platform with free rollers.

Degree of mobility

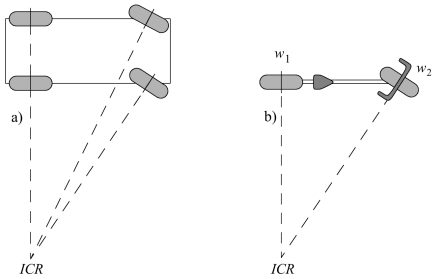


Figure: Wheel axes constrain the instantaneous rotation centre.

How does steering add capability?

$$\delta_m = \dim \mathcal{N}(C_1(\beta_s))$$
$$= 3 - \operatorname{rank} C_1(\beta_s)$$

$\mathcal{N}$: allowed velocities (null space); dim: dimension. Rank counts independent constraints. Steering is held fixed.

Base	Rank	δ_m
Differential	1	2
Ackermann	2	1
Omni	0	3

Steerability and manoeuvrability

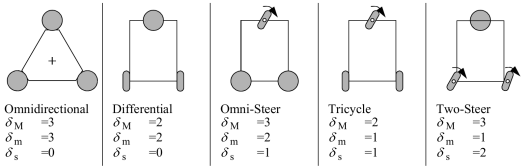


Figure: Five basic wheel arrangements.

Steerability and manoeuvrability

δ_s: number of independent steering freedoms, respecting steering linkages.

$\delta_M = \delta_m + \delta_s$: degree of manoeuvrability.

Base	δ_m	δ_s	δ_M
Differential	2	0	2
Ackermann	1	1	2
Omni	3	0	3

Steering changes allowed motion. Subsequent travel changes the pose.

Which base is easier to park sideways?

Demo 2: park sideways by one metre

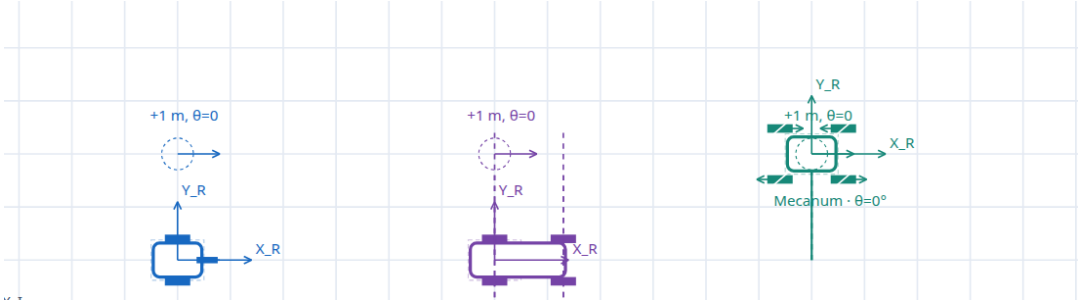


Figure: Only the mecanum base follows a sideways command.

Differential: turn, drive, turn back. Ackermann: forward and reverse arcs. Omni: one straight segment. What should we keep from the lecture?

What to remember

1. Locomotion is a design trade-off

Stability, contact, environment: legs select footholds; wheels suit hard, flat ground.

2. Wheel speeds give velocity; velocity integrates to pose

Use the global and robot frames consistently. Rolling constraints connect wheel rates to motion; sliding constraints restrict it.

3. Mobility and steering are different freedoms

δ_m: instantaneous mobility. δ_s: independent steering. δ_M: their sum.

Next lecture: workspace, holonomic constraints, path versus trajectory, feedback.

Reference Materials

- Siegwart, Nourbakhsh, Scaramuzza — Siegwart Nourbakhsh Scaramuzza Introduction to Autonomous Mobile Robots Second Edition
- Auledas — ANYmal on rough ground, CC BY-SA 4.0
- Daderot — RHex robotic hexapod, CC0
- Jonte — Spot quadruped robot, CC BY-SA 4.0
- GearSkeptic — Rocker-bogie model, CC0
- Anani A. George — Mecanum wheel robot, CC0

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